TOPIC 4

Task 1

```
import Foundation
var subjects = ["Swift", "UIKit", "SwiftUI"]
subjects.append("Networking")
subjects.insert("Testing", at: 1)
subjects.remove(at: subjects.firstIndex(of: "UIKit")!)
print("Sorted Subjects:", subjects.sorted())
var registered: Set<String> = ["Anu", "Ben", "Chitra"]
registered.insert("Dev")
registered.insert("Anu") // Duplicate ignored
print("Registered Students:", registered)
print("Student Count:", registered.count)
var marks: [String: Int] = [
    "Anu": 85,
    "Ben": 90,
    "Chitra": 78
]
marks["Dev"] = 95
marks["Ben"] = 92
print("Esha's Mark:", marks["Esha"] as Any)
for name in marks.keys.sorted() {
    print("\(name): \(marks[name]!)")
}
if let highest = marks.max(by: { $0.value < $1.value }) {
    print("Highest Mark:", highest.value)
    print("Top Student:", highest.key)
}
```

OUTPUT:

```
Sorted Subjects: ["Networking", "Swift", "SwiftUI", "Testing"]
Registered Students: ["Chitra", "Anu", "Ben", "Dev"]
Student Count: 4
Esha's Mark: nil
Anu: 85
Ben: 92
Chitra: 78
Dev: 95
Highest Mark: 95
Top Student: Dev
```

Task 2:

```
import Foundation
let sentence = "swift is fun and swift is fast"
let words = sentence.split(separator: " ").map { String($0) }
print("Total Words:", words.count)
let uniqueWords = Set(words)
print("Unique Words:", uniqueWords.sorted())
var frequency: [String: Int] = [:]
for word in words {
    frequency[word, default: 0] += 1
}
print("Word Frequency:")
for word in frequency.keys.sorted() {
    print("\(word): \(frequency[word]!)")
}
let rankedWords = frequency.sorted {
    if $0.value != $1.value {
        return $0.value > $1.value
    }
```

```
    return $0.key < $1.key
}
print("Ranked Words:")
for item in rankedWords {
    print("\(item.key): \(item.value)")
}
```

OUTPUT:

```
Total Words: 7
Unique Words: ["and", "fast", "fun", "is", "swift"]
Word Frequency:
and: 1
fast: 1
fun: 1
is: 2
swift: 2
Ranked Words:
is: 2
swift: 2
and: 1
fast: 1
fun: 1
```

Task 3:

```
import Foundation
let swiftClub: Set<String> = ["Anu", "Ben", "Chitra", "Dev"]
let uiClub: Set<String> = ["Chitra", "Dev", "Esha"]
let both = swiftClub.intersection(uiClub)
print("Both Clubs:", both.sorted())
let onlySwift = swiftClub.subtracting(uiClub)
print("Only Swift Club:", onlySwift.sorted())
let everyone = swiftClub.union(uiClub)
print("Everyone:", everyone.sorted())
```

```
let exactlyOne = swiftClub.symmetricDifference(uiClub)
print("Exactly One Club:", exactlyOne.sorted())
let numbers = [3, 1, 3, 2, 1, 4]
var seen: Set<Int> = []
var uniqueNumbers: [Int] = []
for number in numbers {
    if seen.insert(number).inserted {
        uniqueNumbers.append(number)
    }
}
print("Original Numbers:", numbers)
print("Without Duplicates:", uniqueNumbers)
```

OUTPUT:

```
Both Clubs: ["Chitra", "Dev"]
Only Swift Club: ["Anu", "Ben"]
Everyone: ["Anu", "Ben", "Chitra", "Dev", "Esha"]
Exactly One Club: ["Anu", "Ben", "Esha"]
Original Numbers: [3, 1, 3, 2, 1, 4]
Without Duplicates: [3, 1, 2, 4]
```